INCH POUND
MIL-PRF-3098/39F
27 August 1997
SUPERSEDING
MIL-C-3098/39E
9 November 1983

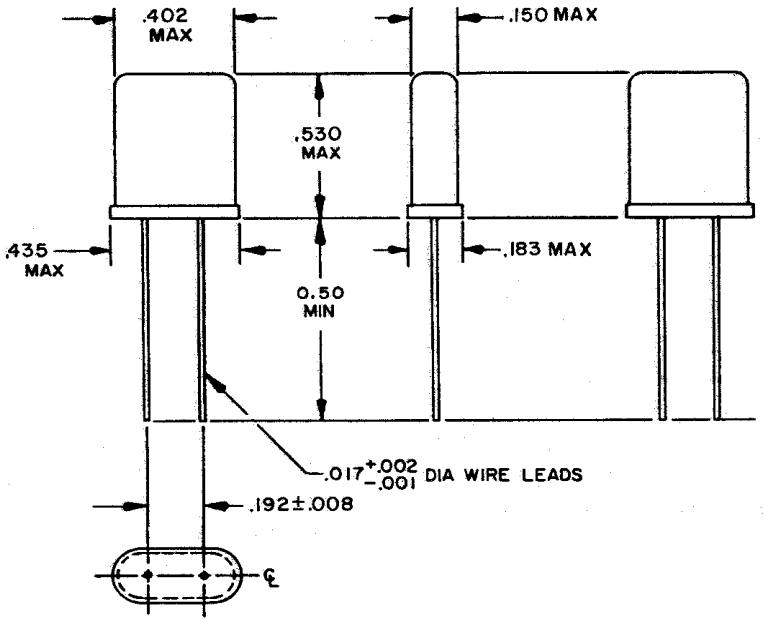
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR61/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 61 MHz; third mechanical overtone; controlled; series resonance.



Inches	mm	Inches	mm	Inches	mm
.001	.03	.150	3.81	.435	11.05
.002	.05	.183	4.65	.500	12.70
.008	.20	.192	4.88	.530	13.46
.017	.43	.402	10.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR-61/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 MHz to 61 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Frequency tolerances:

Operating temperature range: ±20 parts per million (ppm).

Room temperature: ±80 ppm.

Frequency stability: ±5 ppm.

Equivalent resistance: 40 ohms, maximum.

Mode of oscillation: Third mechanical overtone.

Reference temperature: +85°C ±1°C.

Temperature ranges:

Operating (controlled): +80°C to +90°C, inclusive.

Operable: -55°C to +80°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

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Aging:

Frequency change permitted: ±5 ppm.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air force - 85

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 11, 17, 19, 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0697-21)

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